

Introduction to Data Structures and Algorithms



Dr. Sirasit Lochanachit



Outline



1. Resources and Tools
2. What is a “Data Structure”?
3. What is an “Algorithm”?
4. Prerequisites
5. Lesson Plan
6. Grading
7. Suggestions



Resources



Course Materials Website:

https://github.com/noswolf/DSA_AIEE



Tools



1. Google Colaboratory (online)
 - Interactive notebooks (.ipynb)
 - <https://colab.research.google.com/>
 - Login with your kmitl email address
 - [Getting Started with Colab](#)
2. VSCode or any other IDEs (offline)
 - [Getting Started with VSCode](#)



What is a “Data Structure” ?



How do we store, organise, and retrieve data on a computer?



What is a “Data Structure” ?



How do we store, organise, and retrieve data on a computer?

Core concept:

A data structure is a way of **arranging** multiple pieces of information *plus the **rules** for how you're allowed to add, remove, or find things in it.*



Real-life Analogies



Is this a “Data Structure” ?



dreamstime.com

ID 101321023 © Tupungato

Image Source:

<https://www.dreamstime.com/japan-parking-osaka-november-onepark-self-service-automated-plac-es-extremely-popular-congested-japanese-cities-image101321023>

01546203 Data Structures and Algorithms

Is this a “Data Structure” ?

Your friend texts

“My car is in spot 7”.

What will you do to
find your friend's car?



Is this a “Data Structure” ?



Light & Airy Sponge Cake Recipe

★★★★★
5 from 356 votes

Instructions

Sponge Cake

- 1 Preheat oven to 160 °C (320°F) with the fan on (see note 2 if you don't have a fan function) and line the bottom of two 8x3 inch baking tins with baking paper without greasing the bottom. Leave the sides of the cake tins ungreased and bare (do not line with baking paper).
- 2 In a bowl combine your flour, cornstarch, baking powder and salt. Mix until well combined (you do not need to sift it yet). Set aside.
- 3 In a medium sized bowl add in the egg whites and white vinegar, and using a hand or stand mixer, whip on a medium high speed for 30 seconds until foamy. Gradually add in 1/3 cup of the sugar and then continue mixing for a further 2 minutes. Then turn your mixer down to low and mix for a further minute (about 4 minutes of mixing altogether). You should end up with stiff peaks (see note 3). Set aside.



Is this a “Data Structure” ?



Your pajama is in this pile.

How do you find it?



Is this a “Data Structure” ?



Also, can you guess
what are they queuing for?



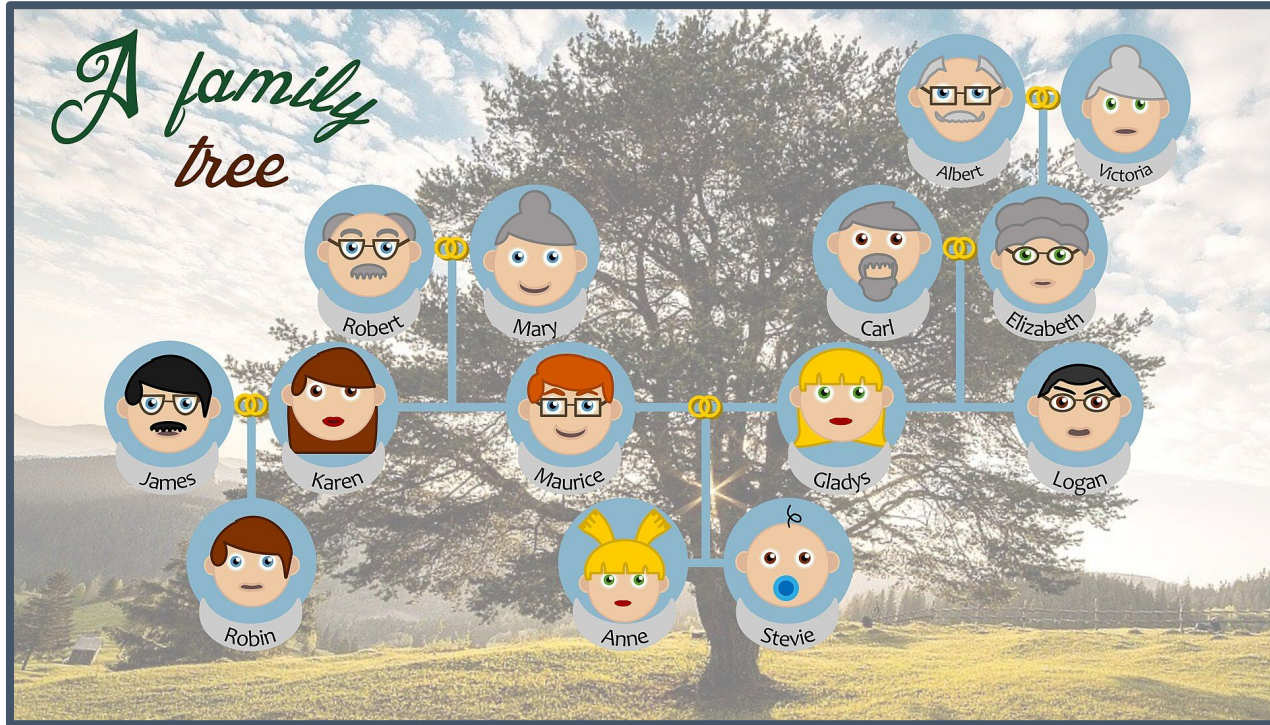
Is this a “Data Structure” ?



What are the rules to get served?

Can you join at the front?

Is this a “Data Structure” ?





Is this a “Data Structure” ?



Address

King Mongkut's Institute of Technology Ladkrabang
1 Chalong Krung 1 Alley, Lat Krabang, Bangkok 10520, Thailand

Phone: 0 2329 8000 - 0 2329 8099

Fax: 0 2329 8106

E-mail

- saraban@kmitl.ac.th
- pr.kmitl@kmitl.ac.th

Is this a “Data Structure” ?



Address

King Mongkut's Institute of Technology Ladkrabang
1 Chalong Krung 1 Alley, Lat Krabang, Bangkok 10520, Thailand

Phone: 0 2329 8000 - 0 2329 8099

Fax: 0 2329 8106

E-mail

- saraban@kmitl.ac.th
- pr.kmitl@kmitl.ac.th

What's KMITL Email address?

How do you find it?



Is this a “Data Structure” ?



Bus No. 1013 originating from Soi Sukhumvit 77 (On Nut) ending at Hua Takhe Market.

Bus No. 1013 (small car) originating from Seri Center Shopping Center Destination Ladkrabang Industrial Estate

Bus No. 517, originating from Chatuchak Park, terminating King Mongkut's Institute of Technology Ladkrabang

Is this a “Data Structure” ?





Is this a “Data Structure” ?



<Buy 1 Free 1> Specialty Pizzas | 485
พิซซ่าสเปเชียล 935
The Truffle Pizza | พืชซ่าทर्फเฟิล 1x
The Maui Wowie | เมออิ วาวอิ (ฮาวายเอี้ยน)



Buy 1 Free 1 | โดนัท 1 แถม 1 130
🇨🇦 Maple Cheesecake | โดนัทไส้เมเปิลชีส 260
Strawberry | แยมสตอร์วเบอร์รี่ 1x
No Sugar | ไม่มีน้ำตาล



What is a “Data Structure” ?



- Way to **store** and **organise** data
 - Can organize many of the same type of things (e.g. cars, people, etc.)
 - Can organize few different types of things that all belong to one entity (e.g. a record, collection of records)
- Enable efficient **access** and **modification** of data



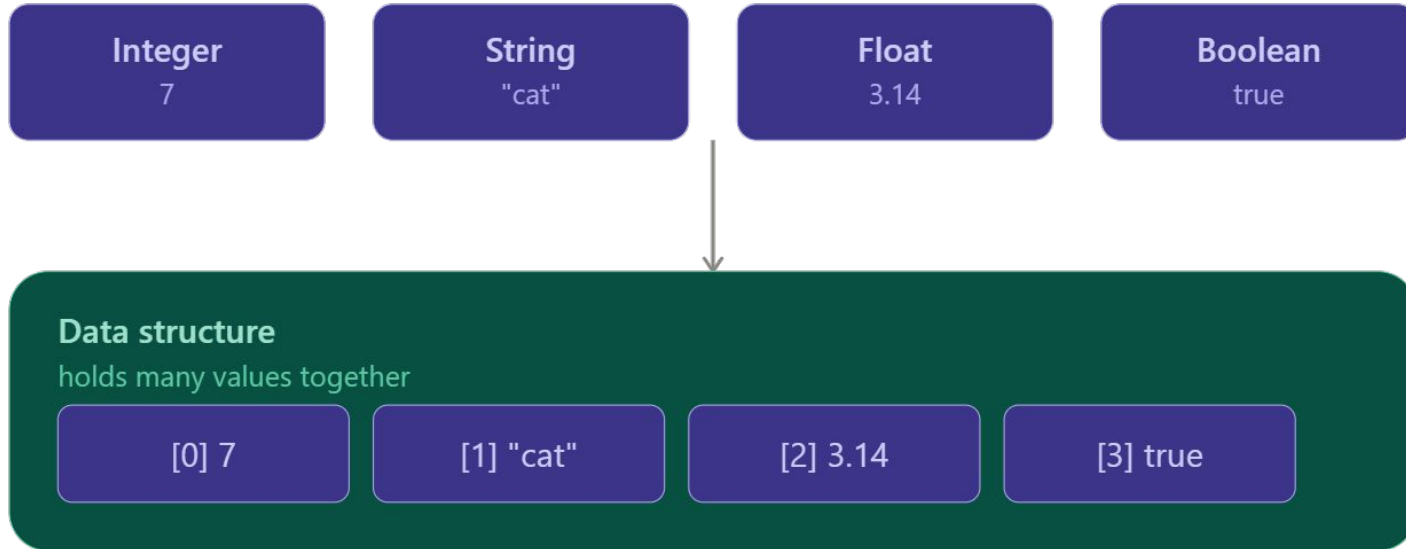
What is a “Data Structure” ?



- Designed for a specific algorithm
 - Strengths and limitations
 - Time and space complexity



Data Type vs Data Structure





Abstract Data Type



- A **data type** tells you what kind of single value something is — like one ingredient in your kitchen
- A **data structure** is how you arrange many of those values together — like the container or shelf you use to organize your ingredients.



Abstract Data Type (ADT)

- ADT is a specification - *what operations are allowed and how the data should behave*, but not about how it is actually built underneath or implemented.
- A data structure is the concrete implementation that actually does the work.



Common vs Abstract Data Type

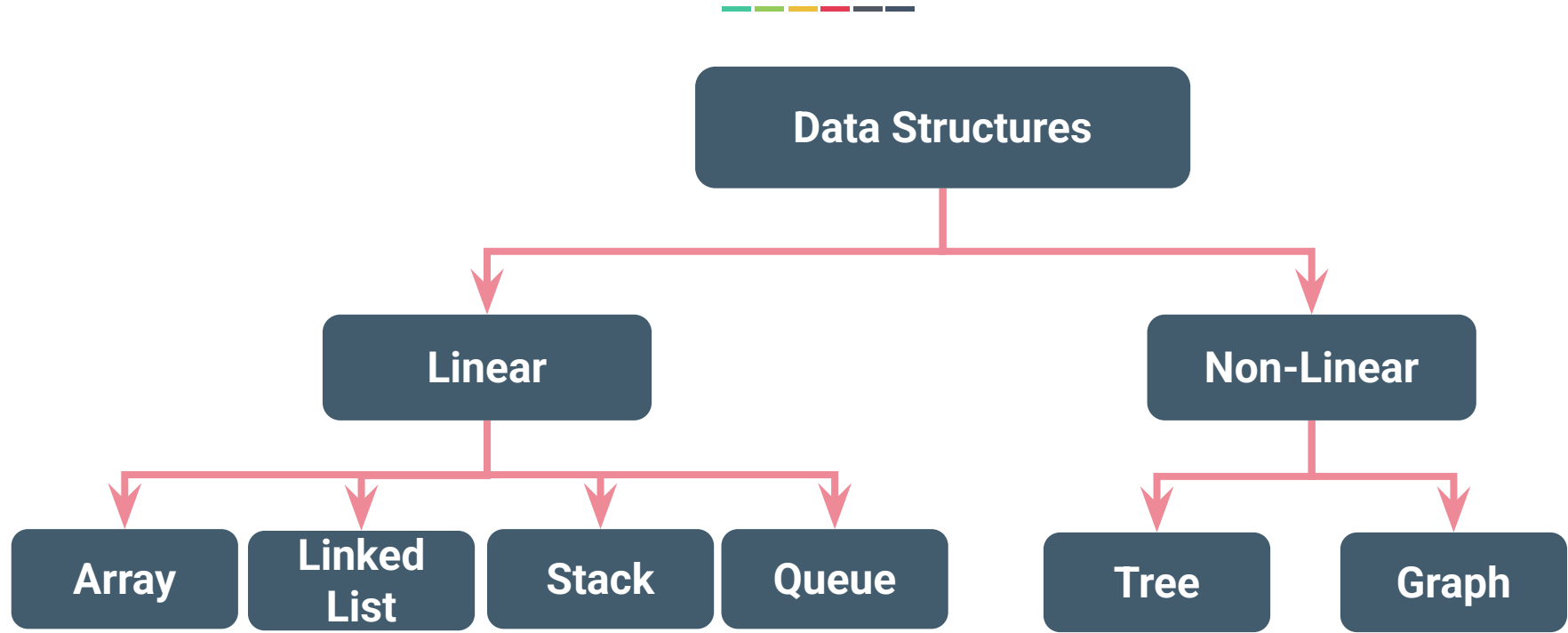
Common

- Integer
- Floating-point number
- Character
- String
- Boolean
- List (for python language)
- etc.

Abstract

- Stack
- Queue
- List (for C language)
- Map
- etc.

Type of Data Structure Covered in this Course



Check out for a comprehensive list of data structures at
https://en.wikipedia.org/wiki/List_of_data_structures



What is an “Algorithm” ?





What is an “Algorithm” ?

- Well-defined procedure or set of instructions to
 - transform input to output or
 - accomplish a task or
 - solve a computational problem





Why care about an “Algorithm” ?

How can we efficiently (in space/time) carry out some typical data processing operations?

How do we analyze and describe their performance?



Example: Sorting numbers





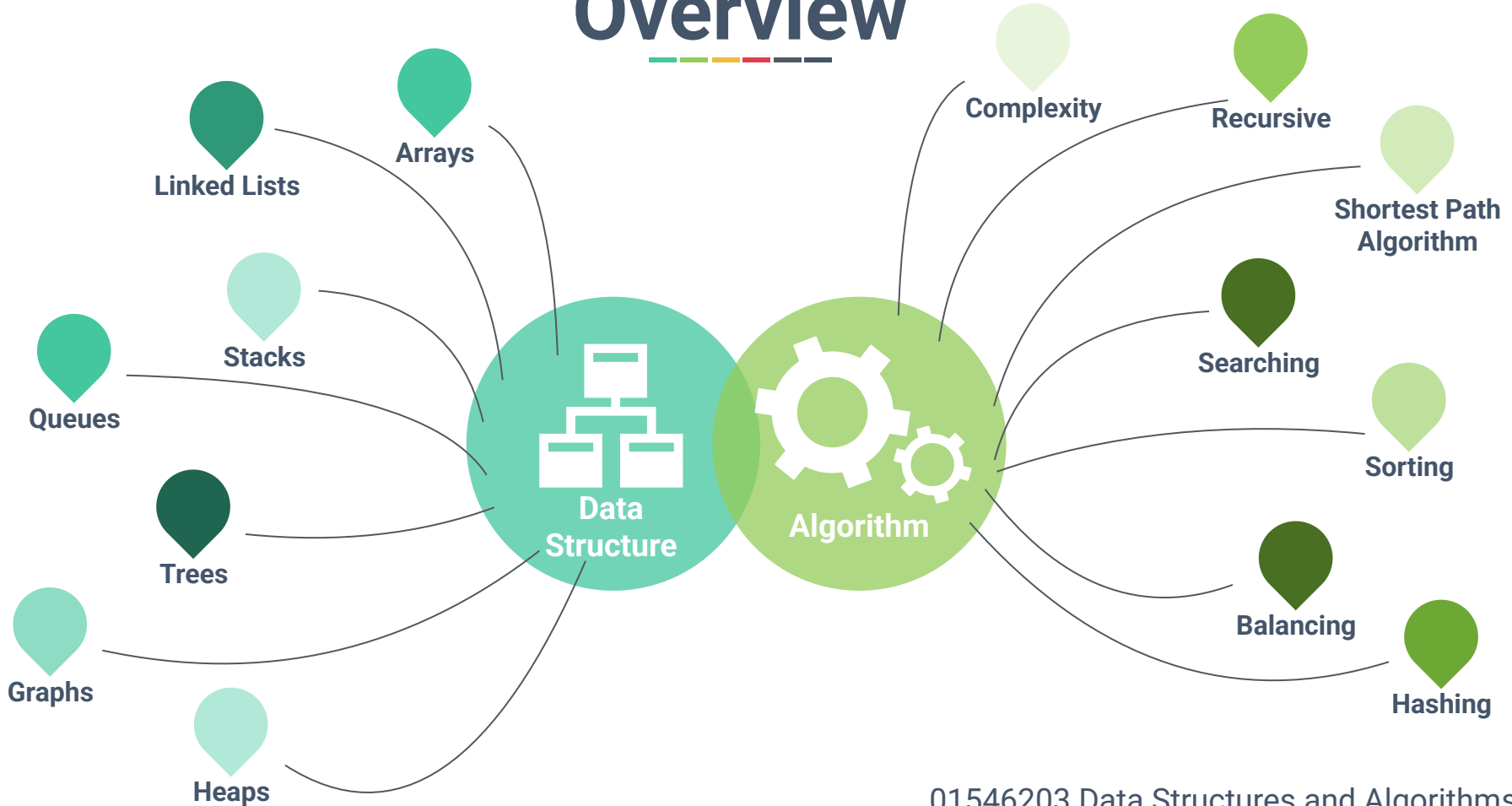
What kind of problems are solved by algorithms?



- Pricing and Valuation Tasks
 - Computing fair prices of derivatives (e.g., options, futures)
- Portfolio Optimization & Asset Allocation
 - Finding the best mix of assets for maximum return at acceptable risk
- Internet: Routing, searches, and security
 - **Shortest path**, search engines, encrypted communication
- E-commerce
 - Ads, recommendations, authentications



Overview





Summary



- Data Structure
 - Way to store and organise data, allowing operations to be performed efficiently.
- Algorithm
 - Step-by-step procedure, which performs on data structure, to be followed to solve a problem/accomplish a task.



Prerequisites

- Fluent in Python Programming
- Comfortable with development processes
 - Writing a function
 - Debugging and testing a code



Lesson Plan

*Public holidays are highlighted in red

Week	Session	Topics	Labs
29/06/2026	01	Introduction	#1
	02	Recursion	#2
06/07/2026	03	Algorithm Analysis	#3
	04	Arrays & OOP	#4
13/07/2026	05	Binary Search	#5
	06	Sorting	#6
20/07/2026	07	Stacks	#7
	08	Queues	#8



Lesson Plan (cont.)

*Public holidays are highlighted in red

Week	Session	Topics	Labs
27/07/2026		Gap Break	
03/08/2026	09	Linked List	#9
	10	Linked List (cont.)	
10/08/2026	11	Trees	#10
	12	Binary Search Trees	
17/08/2026	13	Graphs	#11
	14	Graphs (cont.)	
??/??/2026	15	Final Exam Revision by TA	



Grading

Attendance	5%
Lab Assignment	22%
Midterm Exam	33%
Final Exam	40%



Attendance Score

Out of 13 times, If you attend onsite

12+ times = 5%

11 times = 4%

10 times = 2%

9 times = Unable to take a final exam

Note: Leave notice must be submitted at least 48 hrs before the lecture.

Note 2: Lab submission != Checked Attendance



Lab Assignment

Each lab required to be submitted and presented to TAs during the lab session only

- **Late submission** is not allowed and will not be graded
- Your TAs will be your grader, not your instructor
- Challenge is optional, submission is not required
- If you need any help, feel free to ask me or your TAs
- You can ask your instructor to see a checklist of your submissions.



Suggestions

Working on Lab tasks

- Although this course is intense on programming, you are encouraged to work on lab with your classmates, including having intense discussion.
 - Please do not copy and paste your friend's code.
 - Of course you are welcome to work solo as well.
- If you are stuck, unsure about the task, or have no idea about the error message, feel free to ask TA and they are happy to help.



Suggestions

Practicing with Generative AI

- Even though AI features are allowed and available for you to use freely in this course, please restrain from using it as a first thing when you attempting any task.
- Please Turn off AI generative features in your Colab environment, otherwise you are going to see solutions just as you starting to type.
- It can be very useful if you run out of idea/solution or need a hint from it, including help prepare you for the exam.



Revision at Home

Generative AI (ChatGPT, Gemini) is your go-to.

- Fortunately, most of the responses related to this course would be around 90% correct.
- Purely asking for **What** is the answer for anything is not a true learning.
- The stuff you should ask it:
 - **Why** is it correct/incorrect?
 - **How** does this works?
 - **What if** ?



Things to Seriously Consider

- **Recording** my lectures in any capacity **is not allowed**.
- However, you are allowed to
 - Take photo of my handwritings on the whiteboard or
- Alternatively, Public lectures on YouTube is a great material for you to understand the complex data structures concepts.

Reading List

Essential

Goodrich, M.T., Tamassia, R. and Goldwasser, M.H., 2013. ***Data structures and algorithms in Python***. John Wiley & Sons Ltd.

Recommended

Cormen, T.H., Leiserson, C.E., Rivest, R.L. and Stein, C., 2022. ***Introduction to algorithms***. MIT press.

Miller, B.N. and Ranum, D.L., 2011. ***Problem solving with algorithms and data structures using python***, 2nd ed. Franklin, Beedle & Associates Inc.